

Processing__Italy__SII_demo

October 4, 2025

1 Italian Solvency reports Table S.02.01.02; Part 2 Processing

This script is a continuation of `Extraction_Italy_SII`. It shows the second Phase of this process.

1.1 Companies in scope

For the year 2024, the companies in scope are the following:

- Credemvita S.p.A.
- AXA MPS Assicurazioni Vita
- CRÈDIT AGRICOLE VITA
- Società Reale Mutua di Assicurazioni
- Cardif Vita S.p.A.
- MEDIOLANUM VITA S.p.A.
- Generali Italia S.p.A.
- Banco BPM Vita S.p.A.
- HDI ASSICURAZIONI S.p.A.
- Gruppo Assicurativo Poste Vita
- FIDEURAM VITA S.P.A.
- CNP Vita Assicura S.p.A.
- ITAS VITA
- Helvetia Vita S.p.A.
- Vittoria Assicurazioni S.p.A.
- GROUPAMA ASSICURAZIONI S.P.A.
- UniCredit Allianz Vita S.p.A.
- Zurich Investments Life S.p.A.

1.2 Description of the process

The process of extraction is performed in 5 phases:

1.2.1 Phase 1: Find the reports and identify the relevant tables.

- 1) Identify the new SFCR report and save it into the folder Input.
- 2) Identify the pages where the tables of interest are.
- 3) Compile the map of the company run in the `master_list.csv`.

1.2.2 Phase 2: Run the Extraction script (this script).

The script performs the following steps (with slight modifications depending on the table format):
1) Save the page with the table into a separate folder Single_pdf. 2) Use either a Python package or specialized LLM to create a digital equivalent of the table. 3) Fix the systemic errors that prevent the table from being saved as DataFrame. 4) Save the DataFrame into the Output folder.

1.2.3 Phase 3: Run the Processing script.

The script applies fixes to the DataFrame to make the numbers closer to the reported numbers. It joins all the tables into a single dataset.

1.2.4 Phase 4: Run the Cross-Validation script.

Applies a series of tests that check for the internal consistency between the numbers. Flags the potential errors.

1.2.5 Phase 5: Final modifications to the table and a manual inspection.

1.3 Python packages

```
[929]: import pandas as pd
import numpy as np
```

1.4 Functions

```
[930]: def remove_trailing_zeros(data: pd.DataFrame, col: str, n: int) -> pd.DataFrame:
    """
    Removes exactly n trailing zeros from numbers in a column, if present.

    Args:
        data (pd.DataFrame): Input dataframe.
        col (str): Column name to process.
        n (int): Number of trailing zeros to remove.

    Returns:
        pd.DataFrame: DataFrame with modified column.
    """
    def clean_value(x):
        if pd.isna(x):
            return x
        try:
            # Convert to int first (in case it's float or string with decimals)
            x_int = int(float(x))
            str_x = str(x_int)
            zeros = "0" * n
            if str_x.endswith(zeros):
                return int(str_x[:-n]) # remove trailing zeros
            return x_int
```

```

        except ValueError:
            return x # return unchanged if not a valid number

data[col] = data[col].apply(clean_value)
return data

```

```

[931]: def adjust_decimals(data: pd.DataFrame, col: str = "C0010") -> pd.DataFrame:
        """
        Adjust values in a dataframe column:
        - If a value has non-zero decimal digits, multiply it by 1000.
        - Otherwise, leave it unchanged.

        Parameters
        -----
        data : pd.DataFrame
            Input dataframe with numeric values.
        col : str
            Column name to check (default "C0010").

        Returns
        -----
        pd.DataFrame
            Updated dataframe with adjusted values.
        """
        data = data.copy()
        decimals = data[col] % 1
        data[col] = np.where(decimals != 0, data[col] * 1000, data[col])
        return data

```

```

[932]: def remove_space(data: pd.DataFrame, col: str = "C0010") -> pd.DataFrame:
        """
        Remove every space in a column of strings in a DataFrame.
        """
        data[col] = (
            data[col]
            .str.replace(" ", "", regex=False) # remove thousand separators
        )
        return data

```

```

[933]: def remove_dot_sep(data: pd.DataFrame, col: str = "C0010") -> pd.DataFrame:
        """
        Remove every dot in a column of strings in a DataFrame.
        """
        data[col] = (
            data[col]
            .str.replace(".", "", regex=False)
        )

```

```
return data
```

```
[934]: def as_float(data: pd.DataFrame, col: str = "C0010") -> pd.DataFrame:
        """
        Convert a column of strings in a DataFrame into floating numbers.
        """
        data[col] = (
            data[col]
            .astype(float)
        )
        return data
```

```
[935]: def remove_comma_sep(data: pd.DataFrame, col: str = "C0010") -> pd.DataFrame:
        """
        Remove every comma in a column of strings in a DataFrame.
        """
        data[col] = (
            data[col]
            .str.replace(",", "", regex=False)
        )
        return data
```

```
[936]: def remove_dolar_sep(data: pd.DataFrame, col: str = "C0010") -> pd.DataFrame:
        """
        Remove every dolar sign in a column of strings in a DataFrame.
        """
        data[col] = (
            data[col]
            .str.replace("$", "", regex=False)
        )
        return data
```

```
[937]: def replace_coma_with_dot(data: pd.DataFrame, col: str = "C0010") -> pd.
        ↪DataFrame:
        """
        Replace every comma in a column of strings of a DataFrame with a dot.
        """
        data[col] = (
            data[col]
            .str.replace(",", ".", regex=False)
        )

        return data
```

```
[938]: def fix_dot_numbers(data: pd.DataFrame, col: str) -> pd.DataFrame:
        """
        Fix numbers in a column where dots are used as thousand separators
```

but keep values like '0.000' unchanged.

Args:

data (pd.DataFrame): Input dataframe.

col (str): Column name to fix.

Returns:

pd.DataFrame: DataFrame with the cleaned column.

"""

```
def clean_value(x):
    if pd.isna(x): # Handle NaN
        return x
    x_str = str(x)
    before, _, after = x_str.partition(".")
    if before == "0" and after == "000":
        return x_str # keep '0.000'
    return x_str.replace(".", "") # remove dots otherwise

data[col] = data[col].apply(clean_value)
return data
```

```
[939]: def replace_dash_with_zero(data: pd.DataFrame, col: str) -> pd.DataFrame:
        """
        Replace cells containing only '-' with 0 in the specified column.

        Parameters
        -----
        data : pd.DataFrame
            Input dataframe.
        col : str
            Column name where replacement should happen.

        Returns
        -----
        pd.DataFrame
            DataFrame with '-' replaced by 0 in the specified column.
        """
        data = data.copy()
        data[col] = data[col].apply(lambda x: 0 if str(x).strip() == "-" else x)
        data[col] = data[col].apply(lambda x: 0 if str(x).strip() == "-" else x)
        return data
```

```
[940]: def remove_bracket_instead_negative(data: pd.DataFrame, column: str) -> pd.
        ↪DataFrame:
        """
        Converts numbers stored as strings with brackets into negative numbers.
```

Example: "(123)" -> "-123"

Parameters:

data (pd.DataFrame): Input DataFrame

column (str): Column name where conversion should be applied

Returns:

pd.DataFrame: Updated DataFrame with cleaned column

"""

data = data.copy()

def convert(value):

if isinstance(value, str) and value.startswith("(") and value.
endswith(")"):

return "-" + value[1:-1] # remove brackets, prepend "-"

return value

data[column] = data[column].apply(convert)

return data

```
[941]: def initial_read(unique_id: str, list:pd.DataFrame) -> pd.DataFrame:
```

"""

Find the location of the table and load it.

"""

path = list.loc[unique_id, "output_final_path"]

return pd.read_csv(path, index_col = 0).fillna(0)

```
[942]: def shorten_index(data: pd.DataFrame) -> pd.DataFrame:
```

"""

Trims the index values so they contain only the first 5 characters.

Example:

'R0510 2.337.991' -> 'R0510'

"""

data = data.copy()

data.index = data.index.astype(str).str[:5]

return data

```
[943]: def convert_index_to_R(data: pd.DataFrame) -> pd.DataFrame:
```

"""

Converts the dataframe index by replacing the first digit with 'R'
and formatting the rest as string without decimals.

Example:

80000.0 -> "R0000"

80500.0 -> "R0500"

```

"""
data = data.copy()
data.index = (
    data.index.astype(int).astype(str).str[1:].str.zfill(4).map(lambda x:
↪ "R" + x)
)
return data

```

2 Master list of companies

```
[944]: master_list = pd.read_csv("master_list.csv", header=0, index_col=0)
```

```
[945]: display(master_list)
```

id	company \
CREDEM_VITA_02_1	CREDEM
CREDEM_VITA_02_2	CREDEM
AXA_VITA_02_01	AXA
CREDAG_VITA_02_01	CREDIT_AGRICOLE
CREDAG_VITA_02_02	CREDIT_AGRICOLE
REALE_02_01	REALE_MUTUA
REALE_02_02	REALE_MUTUA
CARDIF_02_01	CARDIF
MEDIO_02_01	MEDIOLANUM
MEDIO_02_02	MEDIOLANUM
GEN_ITA_02_01	GENERALI ITALIA
GEN_ITA_02_02	GENERALI ITALIA
BMP_VITA_02_01	BMP VITA
BMP_VITA_02_02	BMP VITA
HDI_01	HDI
HDI_02	HDI
POSTE_VITA_01	POSTE_VITA
POSTE_VITA_02	POSTE_VITA
INTESA_VITA_01	INTESA_VITA
INTESA_VITA_02	INTESA_VITA
CNP_VITA_01	CNP_VITA
CNP_VITA_02	CNP_VITA
ITAS_VITA_01	ITAS_VITA
ITAS_VITA_02	ITAS_VITA
HELVETIA_VITA_01	HELVETIA_VITA
HELVETIA_VITA_02	HELVETIA_VITA
VITTORIA_01	VITTORIA
VITTORIA_02	VITTORIA
GROUPAMA_01	GROUPAMA
ALLIANZ_UNICREDIT_01	ALLIANZ_UNICREDIT
ALLIANZ_UNICREDIT_02	ALLIANZ_UNICREDIT

ZURICH_LIFE_01
ZURICH_LIFE_02

ZURICH_LIFE
ZURICH_LIFE

id	document_name \
CREDEM_VITA_02_1	Input\SFCR 2024 CREDEMVITA.pdf
CREDEM_VITA_02_2	Input\SFCR 2024 CREDEMVITA.pdf
AXA_VITA_02_01	Input\2024.12 QRT SFCR AXA MPS Assicurazioni V...
CREDAG_VITA_02_01	Input\ca_vita_sfcr_2024.pdf
CREDAG_VITA_02_02	Input\ca_vita_sfcr_2024.pdf
REALE_02_01	Input\SFCR_A123S_20241231.pdf
REALE_02_02	Input\SFCR_A123S_20241231.pdf
CARDIF_02_01	Input\SFCR_A421S_20241231.pdf
MEDIO_02_01	Input\Mediolanum_Vita_Relazione_Unica_2024.pdf
MEDIO_02_02	Input\Mediolanum_Vita_Relazione_Unica_2024.pdf
GEN_ITA_02_01	Input\2024 Relazione sulla Solvibilita e Condi...
GEN_ITA_02_02	Input\2024 Relazione sulla Solvibilita e Condi...
BMP_VITA_02_01	Input\Fascicolo-SFCR_2024_BBPMV-signed_con-opi...
BMP_VITA_02_02	Input\Fascicolo-SFCR_2024_BBPMV-signed_con-opi...
HDI_01	Input\HDI Assicurazioni S.p.A. - Avviso pubbli...
HDI_02	Input\HDI Assicurazioni S.p.A. - Avviso pubbli...
POSTE_VITA_01	Input\relazioneunicasolvibilita_condizionedefina...
POSTE_VITA_02	Input\relazioneunicasolvibilita_condizionedefina...
INTESA_VITA_01	Input\SFCR Gruppo ISPA 2024_ENG.pdf
INTESA_VITA_02	Input\SFCR Gruppo ISPA 2024_ENG.pdf
CNP_VITA_01	Input\SFCR_2024_CNP_Vita_Assicura_con_Relazion...
CNP_VITA_02	Input\SFCR_2024_CNP_Vita_Assicura_con_Relazion...
ITAS_VITA_01	Input\SFCR_G0010_20241231.pdf
ITAS_VITA_02	Input\SFCR_G0010_20241231.pdf
HELVETIA_VITA_01	Input\SFCR_Annex_Helvetia Vita_2024.pdf
HELVETIA_VITA_02	Input\SFCR_Annex_Helvetia Vita_2024.pdf
VITTORIA_01	Input\SFCR-2024_Vittoria.pdf
VITTORIA_02	Input\SFCR-2024_Vittoria.pdf
GROUPAMA_01	Input\SFCR-2024-Relazione-solvibilita-condizio...
ALLIANZ_UNICREDIT_01	Input\UAV_Fascicolo_SFCR_2024_con_relazioni_Pw...
ALLIANZ_UNICREDIT_02	Input\UAV_Fascicolo_SFCR_2024_con_relazioni_Pw...
ZURICH_LIFE_01	Input\zil_scfr_2024_web.pdf
ZURICH_LIFE_02	Input\zil_scfr_2024_web.pdf

id	table_name	page_number \
CREDEM_VITA_02_1	S.02.01.02_A	197
CREDEM_VITA_02_2	S.02.01.02_L	198
AXA_VITA_02_01	S.02.01.02_A	1
CREDAG_VITA_02_01	S.02.01.02_A	61
CREDAG_VITA_02_02	S.02.01.02_L	62
REALE_02_01	S.02.01.02_A	158
REALE_02_02	S.02.01.02_L	159

CARDIF_02_01	S.02.01.02_A	106
MEDIO_02_01	S.02.01.02_A	164
MEDIO_02_02	S.02.01.02_L	165
GEN_ITA_02_01	S.02.01.02_A	144
GEN_ITA_02_02	S.02.01.02_L	145
BMP_VITA_02_01	S.02.01.02_A	80
BMP_VITA_02_02	S.02.01.02_L	81
HDI_01	S.02.01.02_A	104
HDI_02	S.02.01.02_L	105
POSTE_VITA_01	S.02.01.02_A	136
POSTE_VITA_02	S.02.01.02_L	137
INTESA_VITA_01	S.02.01.02_A	250
INTESA_VITA_02	S.02.01.02_L	251
CNP_VITA_01	S.02.01.02_A	116
CNP_VITA_02	S.02.01.02_L	117
ITAS_VITA_01	S.02.01.02_A	233
ITAS_VITA_02	S.02.01.02_L	234
HELVETIA_VITA_01	S.02.01.02_A	2
HELVETIA_VITA_02	S.02.01.02_L	3
VITTORIA_01	S.02.01.02_A	93
VITTORIA_02	S.02.01.02_L	94
GROUPAMA_01	S.02.01.02_A	71
ALLIANZ_UNICREDIT_01	S.02.01.02_A	110
ALLIANZ_UNICREDIT_02	S.02.01.02_L	111
ZURICH_LIFE_01	S.02.01.02_A	58
ZURICH_LIFE_02	S.02.01.02_L	59

output_pdf_path \

id	
CREDEM_VITA_02_1	Single_pdf/CREDEM_S02_01_02_1_2024.pdf
CREDEM_VITA_02_2	Single_pdf/CREDEM_S02_01_02_2_2024.pdf
AXA_VITA_02_01	Single_pdf/AXA_S02_01_02_1_2024.pdf
CREDAG_VITA_02_01	Single_pdf/CA_S02_01_02_1_2024.pdf
CREDAG_VITA_02_02	Single_pdf/CA_S02_01_02_2_2024.pdf
REALE_02_01	Single_pdf/RM_S02_01_02_1_2024.pdf
REALE_02_02	Single_pdf/RM_S02_01_02_2_2024.pdf
CARDIF_02_01	Single_pdf/CARDIF_S02_01_02_1_2024.pdf
MEDIO_02_01	Single_pdf/MEDIO_S02_01_02_1_2024.pdf
MEDIO_02_02	Single_pdf/MEDIO_S02_01_02_2_2024.pdf
GEN_ITA_02_01	Single_pdf/GEN_ITA_S02_01_02_1_2024.pdf
GEN_ITA_02_02	Single_pdf/GEN_ITA_S02_01_02_2_2024.pdf
BMP_VITA_02_01	Single_pdf/BMP_VITA_S02_01_02_1_2024.pdf
BMP_VITA_02_02	Single_pdf/BMP_VITA_S02_01_02_2_2024.pdf
HDI_01	Single_pdf/HDI_S02_01_02_1_2024.pdf
HDI_02	Single_pdf/HDI_S02_01_02_2_2024.pdf
POSTE_VITA_01	Single_pdf/POSTE_VITA_S02_01_02_1_2024.pdf
POSTE_VITA_02	Single_pdf/POSTE_VITA_S02_01_02_2_2024.pdf
INTESA_VITA_01	Single_pdf/INTESA_VITA_S02_01_02_1_2024.pdf

INTESA_VITA_02	Single_pdf/INTESA_VITA_S02_01_02_2_2024.pdf
CNP_VITA_01	Single_pdf/CNP_VITA_S02_01_02_1_2024.pdf
CNP_VITA_02	Single_pdf/CNP_VITA_S02_01_02_2_2024.pdf
ITAS_VITA_01	Single_pdf/ITAS_VITA_S02_01_02_1_2024.pdf
ITAS_VITA_02	Single_pdf/ITAS_VITA_S02_01_02_2_2024.pdf
HELVETIA_VITA_01	Single_pdf/HELVETIA_VITA_S02_01_02_1_2024.pdf
HELVETIA_VITA_02	Single_pdf/HELVETIA_VITA_S02_01_02_2_2024.pdf
VITTORIA_01	Single_pdf/VITTORIA_S02_01_02_1_2024.pdf
VITTORIA_02	Single_pdf/VITTORIA_S02_01_02_2_2024.pdf
GROUPAMA_01	Single_pdf/GROUPAMA_S02_01_02_1_2024.pdf
ALLIANZ_UNICREDIT_01	Single_pdf/ALLIANZ_UNICREDIT_S02_01_02_1_2024.pdf
ALLIANZ_UNICREDIT_02	Single_pdf/ALLIANZ_UNICREDIT_S02_01_02_2_2024.pdf
ZURICH_LIFE_01	Single_pdf/ZURICH_LIFE_S02_01_02_1_2024.pdf
ZURICH_LIFE_02	Single_pdf/ZURICH_LIFE_S02_01_02_2_2024.pdf

	output_final_path \
id	
CREDEM_VITA_02_1	Output/CREDEM_S02_01_02_1_2024.csv
CREDEM_VITA_02_2	Output/CREDEM_S02_01_02_2_2024.csv
AXA_VITA_02_01	Output/AXA_S02_01_02_1_2024.csv
CREDAG_VITA_02_01	Output/CA_S02_01_02_1_2024.csv
CREDAG_VITA_02_02	Output/CA_S02_01_02_2_2024.csv
REALE_02_01	Output/RM_S02_01_02_1_2024.csv
REALE_02_02	Output/RM_S02_01_02_2_2024.csv
CARDIF_02_01	Output/CARDIF_S02_01_02_1_2024.csv
MEDIO_02_01	Output/MEDIO_S02_01_02_1_2024.csv
MEDIO_02_02	Output/MEDIO_S02_01_02_2_2024.csv
GEN_ITA_02_01	Output/GEN_ITA_S02_01_02_1_2024.csv
GEN_ITA_02_02	Output/GEN_ITA_S02_01_02_2_2024.csv
BMP_VITA_02_01	Output/BMP_VITA_S02_01_02_1_2024.csv
BMP_VITA_02_02	Output/BMP_VITA_S02_01_02_2_2024.csv
HDI_01	Output/HDI_S02_01_02_1_2024.csv
HDI_02	Output/HDI_S02_01_02_2_2024.csv
POSTE_VITA_01	Output/POSTE_VITA_S02_01_02_1_2024.csv
POSTE_VITA_02	Output/POSTE_VITA_S02_01_02_2_2024.csv
INTESA_VITA_01	Output/INTESA_VITA_S02_01_02_1_2024.csv
INTESA_VITA_02	Output/INTESA_VITA_S02_01_02_2_2024.csv
CNP_VITA_01	Output/CNP_VITA_S02_01_02_1_2024.csv
CNP_VITA_02	Output/CNP_VITA_S02_01_02_2_2024.csv
ITAS_VITA_01	Output/ITAS_VITA_S02_01_02_1_2024.csv
ITAS_VITA_02	Output/ITAS_VITA_S02_01_02_2_2024.csv
HELVETIA_VITA_01	Output/HELVETIA_VITA_S02_01_02_1_2024.csv
HELVETIA_VITA_02	Output/HELVETIA_VITA_S02_01_02_2_2024.csv
VITTORIA_01	Output/VITTORIA_S02_01_02_1_2024.csv
VITTORIA_02	Output/VITTORIA_S02_01_02_2_2024.csv
GROUPAMA_01	Output/GROUPAMA_S02_01_02_1_2024.csv
ALLIANZ_UNICREDIT_01	Output/ALLIANZ_UNICREDIT_S02_01_02_1_2024.csv
ALLIANZ_UNICREDIT_02	Output/ALLIANZ_UNICREDIT_S02_01_02_2_2024.csv

ZURICH_LIFE_01	Output/ZURICH_LIFE_S02_01_02_1_2024.csv
ZURICH_LIFE_02	Output/ZURICH_LIFE_S02_01_02_2_2024.csv

id	codes_path	leto
CREDEM_VITA_02_1	NaN	2024
CREDEM_VITA_02_2	NaN	2024
AXA_VITA_02_01	NaN	2024
CREDAG_VITA_02_01	NaN	2024
CREDAG_VITA_02_02	NaN	2024
REALE_02_01	NaN	2024
REALE_02_02	NaN	2024
CARDIF_02_01	NaN	2024
MEDIO_02_01	NaN	2024
MEDIO_02_02	NaN	2024
GEN_ITA_02_01	NaN	2024
GEN_ITA_02_02	NaN	2024
BMP_VITA_02_01	NaN	2024
BMP_VITA_02_02	NaN	2024
HDI_01	NaN	2024
HDI_02	NaN	2024
POSTE_VITA_01	NaN	2024
POSTE_VITA_02	NaN	2024
INTESA_VITA_01	NaN	2024
INTESA_VITA_02	NaN	2024
CNP_VITA_01	NaN	2024
CNP_VITA_02	NaN	2024
ITAS_VITA_01	NaN	2024
ITAS_VITA_02	NaN	2024
HELVETIA_VITA_01	NaN	2024
HELVETIA_VITA_02	NaN	2024
VITTORIA_01	NaN	2024
VITTORIA_02	NaN	2024
GROUPAMA_01	NaN	2024
ALLIANZ_UNICREDIT_01	NaN	2024
ALLIANZ_UNICREDIT_02	NaN	2024
ZURICH_LIFE_01	NaN	2024
ZURICH_LIFE_02	NaN	2024

3 Code

3.1 Credemvita S.p.A.

S.01.02.01 1

```
[946]: table_1 = initial_read("CREDEM_VITA_02_1", master_list)
```

```
[947]: table_1 = remove_space(table_1, "C0010")
table_1 = remove_dot_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)
```

S.02.01.02 2

```
[948]: table_2 = initial_read("CREDEM_VITA_02_2", master_list)
```

```
[949]: table_2 = remove_space(table_2, "C0010")
table_2 = remove_dot_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)
```

```
[950]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["CREDEM_VITA"], ["NAME", "C0010"]])
```

```
[951]: credem_02 = table
```

```
[952]: del table, table_1, table_2
```

3.2 AXA MPS Assicurazioni Vita

S.02.01.02

```
[953]: table = initial_read("AXA_VITA_02_01", master_list)
```

```
[954]: table = replace_dash_with_zero(table, "C0010")
table = remove_dot_sep(table, "C0010")
table = replace_coma_with_dot(table, "C0010")
table = remove_dolar_sep(table, "C0010")
table = remove_space(table, "C0010")
table = as_float(table, "C0010")
table = table.fillna(0)
```

```
[955]: table.columns = pd.MultiIndex.from_product([["AXA"], ["C0010"]])
```

```
[956]: axa_02 = table
```

```
[957]: del table
```

3.3 CRÈDIT AGRICOLE VITA

S.02.01.02 1

```
[958]: table_1 = initial_read("CREDAG_VITA_02_01", master_list)
```

```
[959]: table_1 = remove_dot_sep(table_1, "C0010")
table_1 = replace_coma_with_dot(table_1, "C0010")
```

```

table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_space(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)

```

S.02.01.02 2

```
[960]: table_2 = initial_read("CREDAG_VITA_02_02", master_list)
```

```
[961]: table_2 = remove_dot_sep(table_2, "C0010")
table_2 = replace_coma_with_dot(table_2, "C0010")
table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = remove_space(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)

```

```
[962]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.
↳from_product([["CREDIT_AGRICOLE"], ["NAME", "C0010"]])

```

```
[963]: ca_02 = table
```

```
[964]: del table, table_1, table_2
```

3.4 Società Reale Mutua di Assicurazioni

S.02.01.02 1

```
[965]: table_1 = initial_read("REALE_02_01", master_list)
```

S.02.01.02 2

```
[966]: table_2 = initial_read("REALE_02_02", master_list)
```

```
[967]: table_2 = shorten_index(table_2)

table_2 = remove_dot_sep(table_2, "C0010")
table_2 = replace_coma_with_dot(table_2, "C0010")
table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)

```

```
[968]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["REALE_MUTUA"], ["NAME", "C0010"]])

```

```
[969]: rm_02 = table
```

```
[970]: del table, table_1, table_2
```

3.5 Cardif Vita S.p.A.

S.02.01.02

```
[971]: table = initial_read("CARDIF_02_01", master_list)
```

```
[972]: table = table[~table.index.isna()]
```

```
[973]: table = replace_dash_with_zero(table, "C0010")
table = remove_dot_sep(table, "C0010")
table = replace_coma_with_dot(table, "C0010")
table = remove_dolar_sep(table, "C0010")
table = remove_space(table, "C0010")
table = as_float(table, "C0010")
table = table.fillna(0)
```

```
[974]: table.columns = pd.MultiIndex.from_product([["CARDIF"], ["NAME", "C0010"]])
```

```
[975]: cardif_02 = table
```

```
[976]: del table
```

3.6 MEDIOLANUM VITA S.p.A.

S.02.01.02 1

```
[977]: table_1 = initial_read("MEDIO_02_01", master_list)
```

```
[978]: table_1 = remove_dot_sep(table_1, "C0010")
table_1 = remove_space(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)
```

S.02.01.02 2

```
[979]: table_2 = initial_read("MEDIO_02_02", master_list)
```

```
[980]: table_2 = replace_dash_with_zero(table_2, "C0010")
table_2 = remove_space(table_2, "C0010")
table_2 = remove_dot_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)
```

```
[981]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["MEDIOLANUM"], ["NAME", "C0010"]])
```

```
[982]: medio_02 = table
```

```
[983]: del table, table_1, table_2
```

3.7 Generali Italia S.p.A.

S.02.01.02 1

```
[984]: table_1 = initial_read("GEN_ITA_02_01", master_list)
```

```
[985]: table_1 = table_1.loc[table_1.index.notna()]

table_1 = remove_dot_sep(table_1, "C0010")
table_1 = replace_coma_with_dot(table_1, "C0010")
table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_space(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)
```

S.02.01.02 2

```
[986]: table_2 = initial_read("GEN_ITA_02_02", master_list)
```

```
[987]: table_2 = table_2.loc[table_2.index.notna()]

table_2 = remove_dot_sep(table_2, "C0010")
table_2 = replace_coma_with_dot(table_2, "C0010")
table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = remove_space(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)
```

```
[988]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.
    ↪from_product([["GENERALI_ITALIA"], ["NAME", "C0010"]])
```

```
[989]: gen_ita_02 = table
```

```
[990]: del table, table_1, table_2
```

3.8 Banco BPM Vita S.p.A.

S.02.01.02 1

```
[991]: table_1 = initial_read("BMP_VITA_02_01", master_list)
```

```
[992]: table_1 = replace_dash_with_zero(table_1, "C0010")
table_1 = remove_comma_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)
```

S.02.01.02 2

```
[993]: table_2 = initial_read("BMP_VITA_02_02", master_list)
```

```
[994]: table_2 = replace_dash_with_zero(table_2, "C0010")
table_2 = remove_comma_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)
```

```
[995]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["BMP_VITA"], ["NAME", "C0010"]])
```

```
[996]: bmp_vita_02 = table
```

```
[997]: del table, table_1, table_2
```

3.9 HDI ASSICURAZIONI S.p.A.

```
[998]: table_1 = initial_read("HDI_01", master_list)
```

```
[999]: table_1 = remove_space(table_1, "C0010")
table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_dot_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)
```

S.02.01.02 2

```
[1000]: table_2 = initial_read("HDI_02", master_list)
```

```
[1001]: table_2 = table_2[~table_2.index.isna()]
table_2 = table_2[~(table_2.index == '---')]
table_2 = remove_space(table_2, "C0010")
table_2 = remove_dot_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)
```

```
[1002]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["HDI"], ["NAME", "C0010"]])
```

```
[1003]: hdi_02 = table
```

```
[1004]: del table, table_1, table_2
```

3.10 Gruppo Assicurativo Poste Vita

S.02.01.02 1

```
[1005]: table_1 = initial_read("POSTE_VITA_01", master_list)
```

```
[1006]: table_1 = remove_space(table_1, "C0010")
table_1 = replace_dash_with_zero(table_1, "C0010")
```



```

table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_bracket_instead_negative(table_1, "C0010")
table_1 = remove_dot_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)

```

S.02.01.02 2

```
[1007]: table_2 = initial_read("POSTE_VITA_02", master_list)
```

```
[1008]: table_2 = table_2.loc[table_2.index!="-"]

table_2 = remove_space(table_2, "C0010")
table_2 = replace_dash_with_zero(table_2, "C0010")
table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = remove_bracket_instead_negative(table_2, "C0010")
table_2 = remove_dot_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)

```

```
[1009]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["POSTE_VITA"], ["NAME", "C0010"]])
```

```
[1010]: poste_vita_02 = table
```

```
[1011]: del table, table_1, table_2
```

3.11 FIDEURAM VITA S.P.A.

S.02.01.02 1

```
[1012]: table_1 = initial_read("INTESA_VITA_01", master_list)
```

```
[1013]: table_1 = table_1.loc[table_1.index.notna()]

table_1 = replace_dash_with_zero(table_1, "C0010")
table_1 = remove_space(table_1, "C0010")
table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_comma_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)

```

S.02.01.02 2

```
[1014]: table_2 = initial_read("INTESA_VITA_02", master_list)
```

```
[1015]: table_2 = table_2.loc[table_2.index.notna()]

table_2 = replace_dash_with_zero(table_2, "C0010")

```

```

table_2 = remove_space(table_2, "C0010")
table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = remove_comma_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)

```

```

[1016]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["INTESA_VITA"], ["NAME", "C0010"]])

```

```

[1017]: intesa_vita_02 = table

```

```

[1018]: del table, table_1, table_2

```

3.12 CNP Vita Assicura S.p.A.

S.02.01.02 1

```

[1019]: table_1 = initial_read("CNP_VITA_01", master_list)

```

```

[1020]: table_1 = table_1[~table_1.index.isna()]
table_1 = convert_index_to_R(table_1)

table_1 = replace_dash_with_zero(table_1, "C0010")
table_1 = remove_space(table_1, "C0010")
table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_dot_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)

```

S.02.01.02 2

```

[1021]: table_2 = initial_read("CNP_VITA_02", master_list)

```

```

[1022]: table_2 = convert_index_to_R(table_2)

table_2 = replace_dash_with_zero(table_2, "C0010")
table_2 = remove_space(table_2, "C0010")
table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = remove_dot_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)

```

```

[1023]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["CNP_VITA"], ["NAME", "C0010"]])

```

```

[1024]: cnp_vita_02 = table

```

```

[1025]: del table, table_1, table_2

```

3.13 ITAS VITA

S.02.01.02 1

```
[1026]: table_1 = initial_read("ITAS_VITA_01", master_list)
```

```
[1027]: table_1 = table_1.loc[table_1.index.notna()]

table_1 = replace_dash_with_zero(table_1, "C0010")
table_1 = remove_space(table_1, "C0010")
table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_dot_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)
```

S.02.01.02 2

```
[1028]: table_2 = initial_read("ITAS_VITA_02", master_list)
```

```
[1029]: table_2 = table_2[~table_2.index.isna()]

table_2 = replace_dash_with_zero(table_2, "C0010")
table_2 = remove_space(table_2, "C0010")
table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = remove_dot_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)
```

```
[1030]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["ITAS_VITA"], ["NAME", "C0010"]])
```

```
[1031]: itas_vita_02 = table
```

```
[1032]: del table, table_1, table_2
```

3.14 Helvetia Vita S.p.A.

S.02.01.02 1

```
[1033]: table_1 = initial_read("HELVETIA_VITA_01", master_list)
```

```
[1034]: table_1 = replace_dash_with_zero(table_1, "C0010")
table_1 = remove_space(table_1, "C0010")
table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_dot_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)
```

S.02.01.02 2

```
[1035]: table_2 = initial_read("HELVETIA_VITA_02", master_list)

[1036]: table_2 = replace_dash_with_zero(table_2, "C0010")
table_2 = remove_space(table_2, "C0010")
table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = remove_dot_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)

[1037]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["HELVATIA_VITA"], ["NAME", "C0010"]])

[1038]: helvatia_vita_02 = table

[1039]: del table, table_1, table_2
```

3.15 Vittoria Assicurazioni S.p.A.

S.02.01.02 1

```
[1040]: table_1 = initial_read("VITTORIA_01", master_list)

[1041]: table_1 = replace_dash_with_zero(table_1, "C0010")
table_1 = remove_space(table_1, "C0010")
table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_bracket_instead_negative(table_1, "C0010")
table_1 = remove_dot_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)
```

S.02.01.02 2

```
[1042]: table_2 = initial_read("VITTORIA_02", master_list)

[1043]: table_2 = replace_dash_with_zero(table_2, "C0010")
table_2 = remove_space(table_2, "C0010")
table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = remove_bracket_instead_negative(table_2, "C0010")
table_2 = remove_dot_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)

[1044]: table = pd.concat([table_1, table_2])
table.columns = pd.MultiIndex.from_product([["VITTORIA"], ["C0010"]])

[1045]: vittora_vita_02 = table

[1046]: del table, table_1, table_2
```

3.16 GROUPAMA ASSICURAZIONI S.P.A.

S.02.01.02 1

```
[1047]: table = initial_read("GROUPAMA_01", master_list)
```

```
[1048]: table = table[~table.index.isna()]
table = table[~(table.index == '---')]
```

```
[1049]: table = replace_dash_with_zero(table, "C0010")
table = remove_space(table, "C0010")
table = remove_dolar_sep(table, "C0010")
table = remove_bracket_instead_negative(table, "C0010")
table = remove_dot_sep(table, "C0010")
table = as_float(table, "C0010")
table = table.fillna(0)
```

```
[1050]: table.columns = pd.MultiIndex.from_product([["GROUPAMA"], ["NAME", "C0010"]])
```

```
[1051]: groupama_02 = table
```

```
[1052]: del table
```

3.17 UniCredit Allianz Vita S.p.A.

S.02.01.02 1

```
[1053]: table_1 = initial_read("ALLIANZ_UNICREDIT_01", master_list)
```

```
[1054]: table_1 = table_1.loc[table_1.index.notna()]

table_1 = replace_dash_with_zero(table_1, "C0010")
table_1 = remove_space(table_1, "C0010")

table_1 = remove_dolar_sep(table_1, "C0010")
table_1 = remove_bracket_instead_negative(table_1, "C0010")
table_1 = remove_dot_sep(table_1, "C0010")
table_1 = as_float(table_1, "C0010")
table_1 = table_1.fillna(0)
```

S.02.01.02 2

```
[1055]: table_2 = initial_read("ALLIANZ_UNICREDIT_02", master_list)
```

```
[1056]: table_2 = table_2.loc[table_2.index.notna()]

table_2 = replace_dash_with_zero(table_2, "C0010")
table_2 = remove_space(table_2, "C0010")
```

```

table_2 = remove_dolar_sep(table_2, "C0010")
table_2 = remove_bracket_instead_negative(table_2, "C0010")
table_2 = remove_dot_sep(table_2, "C0010")
table_2 = as_float(table_2, "C0010")
table_2 = table_2.fillna(0)

```

```

[1057]: table = pd.concat([table_1, table_2])
        table.columns = pd.MultiIndex.
           ↪from_product([["ALLIANZ_UNICREDIT"], ["NAME", "C0010"]])

```

```

[1058]: allianz_02 = table

```

```

[1059]: del table, table_1, table_2

```

3.18 Zurich Investments Life S.p.A.

```

[1060]: table_1 = initial_read("ZURICH_LIFE_01", master_list)

```

```

[1061]: table_1 = replace_dash_with_zero(table_1, "C0010")
        table_1 = remove_space(table_1, "C0010")
        table_1 = as_float(table_1, "C0010")
        table_1 = table_1.fillna(0)

```

S.02.01.02 2

```

[1062]: table_2 = initial_read("ZURICH_LIFE_02", master_list)

```

```

[1063]: table_2 = replace_dash_with_zero(table_2, "C0010")
        table_2 = table_2.fillna(0)

```

```

[1064]: table = pd.concat([table_1, table_2])
        table.columns = pd.MultiIndex.from_product([["ZURICH_LIFE"], ["NAME", "C0010"]])

```

```

[1065]: zurich_life_02 = table

```

```

[1066]: del table, table_1, table_2

```

3.19 Dirty master table

```

[1067]: master_ita_02 = pd.read_csv("Input/S02_01_02_Master_Table.csv", index_col = 0).
           ↪fillna(0)
        master_ita_02.columns = pd.MultiIndex.from_product([["MASTER"], ["NAME"]])

```

```

[1068]: credem_02 = credem_02.drop(columns=("CREDEM_VITA", "NAME"))
        #axa_02 = axa_02.drop(columns=("AXA", "NAME"))
        ca_02 = ca_02.drop(columns=("CREDIT_AGRICOLE", "NAME"))
        rm_02 = rm_02.drop(columns=("REALE_MUTUA", "NAME"))
        cardif_02 = cardif_02.drop(columns=("CARDIF", "NAME"))

```

```

medio_02 = medio_02.drop(columns=("MEDIOLANUM", "NAME"))
gen_ita_02 = gen_ita_02.drop(columns=("GENERALI_ITALIA", "NAME"))
bmp_vita_02 = bmp_vita_02.drop(columns=("BMP_VITA", "NAME"))
hdi_02 = hdi_02.drop(columns=("HDI", "NAME"))
poste_vita_02 = poste_vita_02.drop(columns=("POSTE_VITA", "NAME"))
intesa_vita_02 = intesa_vita_02.drop(columns=("INTESA_VITA", "NAME"))
cnp_vita_02 = cnp_vita_02.drop(columns=("CNP_VITA", "NAME"))
itas_vita_02 = itas_vita_02.drop(columns=("ITAS_VITA", "NAME"))
helvatia_vita_02 = helvatia_vita_02.drop(columns=("HELVATIA_VITA", "NAME"))
#vittoria_vita_02 = vittoria_vita_02.drop(columns=("VITTORIA", "NAME"))
groupama_02 = groupama_02.drop(columns=("GROUPAMA", "NAME"))
allianz_02 = allianz_02.drop(columns=("ALLIANZ_UNICREDIT", "NAME"))
zurich_life_02 = zurich_life_02.drop(columns=("ZURICH_LIFE", "NAME"))

```

```

[1069]: table_02 = master_ita_02.join(credem_02).join(axa_02).join(ca_02).join(rm_02).
    ↪join(cardif_02).join(medio_02).join(gen_ita_02).join(bmp_vita_02).
    ↪join(hdi_02).join(poste_vita_02).join(intesa_vita_02).join(cnp_vita_02).
    ↪join(itas_vita_02).join(helvatia_vita_02).join(vittoria_vita_02).
    ↪join(groupama_02).join(allianz_02).join(zurich_life_02)

```

```

[1070]: table_02 = table_02[~table_02.index.duplicated(keep=False)]

```

```

[1071]: table_02.to_csv("Dirty_Combined/Table_ita_S02.csv")

```